

Token Cost and Early-Termination Break-Even Analysis for Scopey

A controlled component evaluation of local Qwen3.5 9B Q4

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Abstract

We extend the SCOPEY scope-transition benchmark with explicit token accounting. The experiment compares a no-SCOPEY control with the selected local summarizer, Qwen3.5 9B Q4 served by llama.cpp, over 12 synthetic scenarios, 22 scored turns, and three repetitions. Per repetition, the measured component workload contains 431 main-session user-context tokens, 7,958 SCOPEY analyzer-input tokens, and 1,613 SCOPEY-generated tokens. Total analyzer overhead is 9,571 tokens, or 435 tokens per scored turn. Because the component corpus contains no main-agent generations or tool transcripts, early-termination savings are not observed. Instead, we report the falsifiable break-even condition: an intervention must prevent an average of **798 future main-session tokens per scenario** to recover analyzer overhead. If one intervention prevents a 2,500-token suffix in each scenario, the projected reduction is **20,429 tokens per corpus repetition (67.1%)**. This projection is a sensitivity result, not a causal end-to-end measurement.

1 Question and experimental boundary

The practical question is whether the tokens spent maintaining an explicit task scope are repaid when SCOPEY stops an agent from continuing down an obsolete or off-track trajectory. The accounting separates three quantities:

1. **Main-session tokens** (M): cumulative user-request context in the component fixture. Main-agent output, reasoning, tool calls, and tool results are absent and therefore are not counted.
2. **Scopey analyzer input** (S_i): tokens submitted to the local scope-state transformer, including its instructions, prior scope, recent requests, and latest request.
3. **Scopey-generated tokens** (S_o): completion tokens produced by the local model. Total analyzer overhead is $H = S_i + S_o$.

This boundary matters. The evaluation measures summarization cost exactly for the chosen local runtime, but it does not measure a full coding-agent session. Consequently, avoided future work is expressed as a parameter rather than silently inferred from synthetic scope text.

2 Method

2.1 Configurations

The control exposes only the latest user prompt and performs no model-backed scope analysis. The treatment uses the selected local Qwen3.5 9B Q4 GGUF model through a warm llama.cpp server. Decoding uses temperature zero, seed 42, a 256-token completion limit, and validated JSON normalized into SCOPEY’s scope format. The runtime revision is `fe2adf0e722f30f5295fdec8a0f1dc788f7498bc`; the model SHA-256 is `d784ce9eda1a5a7b51e8f705a9e6310844bf4f173654d115823c775fdea56d43`.

The 12 cases exercise additions, narrowing, subtraction, explicit and implicit replacement, read-only queries, administration, combined mutations, constraint preservation, non-invention, and reference resolution. Each run contains 22 scored scope transitions. Three repetitions produce 66 samples per configuration and 132 samples overall.

2.2 Counting and projection

Qwen tokenization is used for main-session fixture text. The llama.cpp OpenAI-compatible response supplies analyzer prompt and completion usage, so the SCOPEY counts are runtime-reported rather than byte estimates. Counts in Tables 1–2 are normalized to one corpus repetition.

Let $N = 12$ scenarios and let A be the future main-session tokens prevented by one successful early termination in each scenario. The modeled totals are

$$T_{\text{no scopey}}(A) = M + NA, \tag{1}$$

$$T_{\text{scopey}} = M + H, \tag{2}$$

$$\Delta(A) = NA - H. \tag{3}$$

Positive Δ indicates net token savings. The corpus breaks even at $A = H/N = 9,571/12 = 797.6$ tokens per scenario. Scenario-specific break-even values use each scenario’s measured analyzer overhead.

3 Results

Table 1: Aggregate token accounting per 12-scenario corpus repetition.

Token bucket	No SCOPEY	Local SCOPEY
Main-session user context (M)	431	431
Analyzer input (S_i)	0	7,958
SCOPEY-generated output (S_o)	0	1,613
Total analyzer overhead (H)	0	9,571
Observed component total	431	10,002

Analyzer input accounts for 83.1% of SCOPEY overhead and generated output for 16.9%. The model generates 73.3 tokens per scored turn on average; full analyzer cost is 435.0 tokens per turn. The no-SCOPEY control has no analyzer cost. Without an avoided future trajectory, it is necessarily cheaper in token terms; SCOPEY’s value appears only when a correction prevents enough later main-session work.

Table 2: Per-scenario token cost and break-even, averaged over three repetitions. “Main” is cumulative user-request context. “Generated” is model output.

Scenario	Turns	Main	Input	Gen.	Cost	Net at 2.5k
Initial concrete addition	1	15	352	65	417	2,083
Follow-up preserving work	2	32	707	120	827	1,673
Narrow target	2	42	738	147	885	1,615
Subtract requirement	2	44	738	143	881	1,619
Explicit task replacement	2	47	726	157	883	1,617
Implicit topic switch	2	40	717	140	857	1,643
Query preserves work	2	38	715	147	862	1,638
Administration preserves work	2	31	713	141	854	1,646
Combined subtract/add	2	47	737	158	895	1,605
Cross-platform constraints	2	51	757	201	958	1,542
No invented administration	1	10	347	53	400	2,100
Reference resolution	2	34	711	141	852	1,648
Corpus total	22	431	7,958	1,613	9,571	20,429

3.1 Early-termination sensitivity

Table 3 makes the counterfactual assumption visible. At zero avoided tokens, SCOPEY adds its full 9,571-token analyzer cost. Savings turn positive near 798 tokens per scenario. At the requested illustrative 2,500-token suffix, the no-SCOPEY continuation would consume 30,431 tokens, whereas the terminated SCOPEY path consumes 10,002, a 67.1% reduction.

Table 3: Sensitivity to assumed future main-session tokens prevented per scenario.

Avoided per scenario (A)	Net corpus savings $\Delta(A)$	Reduction
0	-9,571	-2,220.6%
250	-6,571	-191.5%
500	-3,571	-55.5%
800	29	0.3%
1,000	2,429	19.5%
2,500	20,429	67.1%
5,000	50,429	83.4%

The local model retained the earlier quality result during this run: 100% format compliance, 100% required-concept recall, 100% forbidden-concept rejection, zero call failures, and 59.1% exact transition-label agreement. Median latency was 1,190 ms. The lower transition-label score reflects a coarser operation taxonomy; the active scope body consumed by the product remained correct on all measured concepts.

4 Threats to validity

- **No causal termination measurement.** The 2,500-token suffix is a sensitivity parameter. No main agent was actually stopped in this component experiment.
- **Partial main-session accounting.** The 431-token main bucket contains user-request context only. It omits system prompts, assistant output, reasoning, cache reads/writes, and tool payloads.
- **One tokenizer.** Counts use the selected Qwen tokenizer. Cloud providers may tokenize identical text differently.

- **Synthetic cases.** The corpus is deterministic and auditable but is smaller and cleaner than long production coding sessions.
- **Intervention frequency.** The projection assumes one successful termination per scenario. Real prevalence and false-positive rates determine realized savings.

5 Conclusion and next experiment

Local scope summarization is not token-free: the selected configuration costs about 435 analyzer tokens per scored turn and 798 tokens per scenario. It is token-positive only when a correction prevents more future main-session work than that overhead. Under a 2,500-token prevented-suffix scenario, the modeled benefit is substantial, but it remains projected.

The next evaluation should run paired disposable coding tasks with matched model, prompt, seed, repository state, and stopping rule. It must capture provider-reported input, output, cached, and reasoning tokens at every main-agent step; record the precise intervention token index; continue the no-SCOPEY arm to completion; and score task success plus false interventions. That experiment would replace *A* with observed counterfactual trajectory differences and turn this break-even analysis into an end-to-end causal estimate.

Reproducibility

The evaluator, case corpus, run command, and machine-readable schema live under `eval/`. The source paper is `eval/paper/scopey-token-accounting.tex`. The selected model and runtime are documented in `eval/LOCAL_MODEL_DECISION.md`. Runtime and model projects: <https://github.com/ggml-org/llama.cpp> and <https://huggingface.co/Qwen/Qwen3.5-9B>.